

8(a) (i)	It represents the <u>rated</u> power output that the turbine is expected to generate under <u>ideal</u> operating conditions. The actual power output can vary based on factors such as wind speed and availability, air density, turbulence, temperature, altitude, and other environmental conditions.	B1
8(a) (ii)	The energy <u>demand</u> or consumption of homes can <u>vary</u> significantly depending on factors such as the size of the house, the number of occupants, energy-efficient appliances, and individual usage patterns.	B1
8(b)	Minimum height = $56 - 44/2 = 34$ m	A1
8(c) (i)1	600 kW	A1
8(c) (i)2	15.0 m s^{-1}	A1
8(c) (ii)1	$E_{\max} = \frac{1}{2} \pi r^2 v^3 \rho = \frac{1}{2} \pi \left(\frac{44}{2} \right)^2 (15.0)^3 (1.25)$ $= 3.21 \times 10^6 \text{ W}$	C1 A1

8(c) (ii)2	Overall efficiency = $\frac{P_{out}}{P_{in}} \times 100\% = \frac{600 \times 10^3}{3.21 \times 10^6} \times 100\%$ = 18.7%	C1 A1
8(c) (iii)	$P_{remaining} = P_{in} - P_{out} - 10\%P_{in}$ = 2.29×10^6 W (3 s.f.)	C1 A1
8(c) (iv)1	25 m s ⁻¹	A1
8(c) (iv)2	A safety measure: it helps <u>avoid overloading of the electrical system</u> and generator, <u>preventing equipment damage</u> or electrical failures. OR At high wind speeds, the forces acting on the blades can become <u>excessive</u> , potentially causing <u>damage</u> such as blade failure, gearbox stress, or tower vibrations. [If the cut-out mechanism also involves shutting down the turbine and feather the blades – turn them out of the wind) to prevent over-rotation.]	B1 [B1]
8(d) (i)	Alternating current because in Fig. 8.3, it shows a frequency of 50 Hz. (Accept explanation using A.C. generator concept: e.g., as the hub and blades rotate, they drive the generator's rotor, which contains magnets or magnetic fields. This rotor is mounted on a shaft and rotates within proximity to a stationary coil. This relative motion results in a dynamic change in magnetic flux linkage, which, in turn, induces an e.m.f. in the coil. This induced voltage takes the form of AC because the rotor's rotation periodically alters the magnetic field's orientation relative to the coil.)	B1 B1
8(d) (ii)	$P_{nominal} = I_{nominal}V$ $I_{nominal} = \frac{500 \times 10^3}{690}$ = 725 A (3 s.f.)	C1 A1
8(e) (i)	The <u>rotor blades</u> of a wind turbine are typically the <u>highest</u> and most exposed part of the entire structure, and most likely to be struck by lightning. Their height and exposure, positioning, blade geometry (e.g., pointed tips) make them more susceptible.	B1 B1
8(e) (ii)	Installing a <u>lightning protection system</u> (lightning rods in blades to attract lightning strikes) and <u>proper grounding</u> (connecting metal parts of the turbine with conductive cables or grounding systems) to create low resistance path to dissipate the lightning current to the ground. Additional information: Wind turbine blades are commonly made using composite materials (e.g., fiberglass-reinforced composites) that are non-	B1 B1

DUNMAN HIGH SCHOOL
2023 H2 PHYSICS (YEAR 6)

	conductive. Manufacturers must often incorporate lightning protection systems into the blade design to mitigate the risk associated with lightning strikes.	
--	---	--